GRADE 7 ACCELERATED · UNIT 6 · LESSONS 1–2

Triangle Inequality Theorem; Converse of the Pythagorean Theorem

Independent Set 6.01 **ANSWER KEY**

Name
Date
Period

Six questions, on your own, in silence. Show the step that does the work.

1. Can sides of 4 cm, 8 cm and 13 cm make a triangle?

No — 4 + 8 = 12 is less than 13.

2. Can sides of 9 inches, 9 inches and 18 inches make a triangle?

No — 9 + 9 = 18 lies flat.

The equal case is not a triangle.

3. Two sides of a triangle are 5 m and 11 m. Write an inequality for the length x, in meters, of the third side.

6 < x < 16

11 − 5 and 11 + 5; never equal.

4. Do sides of 28 cm, 45 cm and 53 cm make a right triangle?

Yes — 784 + 2,025 = 2,809 = 53^2.

5. Do sides of 17 ft, 10 ft and 14 ft make a right triangle?

No — $10^2 + 14^2 = 296$, but $17^2 = 289$.

The longest side is listed first.

6. A carpenter joins boards of 30 inches, 40 inches and 50 inches into a triangle. Is one corner a right angle?

Yes — 900 + 1,600 = 2,500 = 50^2.